# A Problem with Presidents

Zlata Kovrigina · October 30, 2026

## Summary

Of the 45 presidents in the data set, 39 had died. They lived 26,097 days (71.5 years) on average, with a median of 70.3 years and a standard deviation of 12.6 years. George Bush lived longest (94 years) and John F. Kennedy shortest (46 years).

## Methods and assumptions

Data: the CSV from the challenge repository (PresidentsUSA.Net, collected September 28, 2021), used exactly as given. The last row is a citation, so it was removed.

Date math: pandas and dateutil's relativedelta count real calendar years and months, so leap years and month lengths are correct. lived_years = completed years, lived_months = total completed months, lived_days = exact days from birth to death.

Living presidents (Jimmy Carter, Bill Clinton, George W. Bush, Barack Obama, Donald Trump, Joe Biden) have no death date. Their ages are measured to the date the data was collected. Their lifespans aren't finished, so they are left out of the rankings and statistics.

Weighted mean: the challenge doesn't say what to weight by, and equal weights just give the mean. Each president is weighted by their position in the CSV (first row = 1), so later presidents count more.

Mode: no two presidents lived the same number of days, so there is no mode.

Standard deviation: sample version (divides by n - 1).

## Top 10 longest-lived presidents

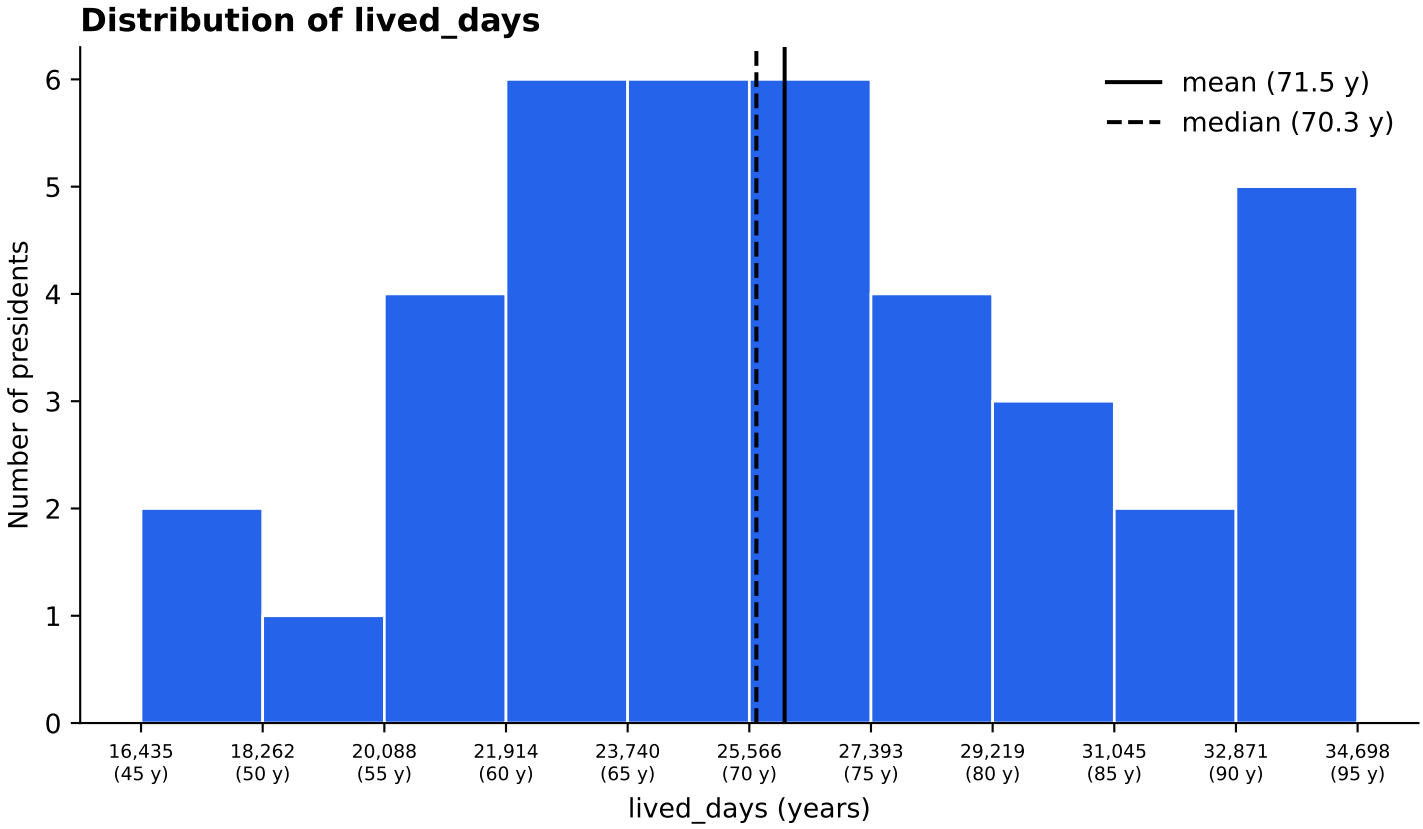
| rank | PRESIDENT | year_of_birth | lived_years | lived_months | lived_days |
|---|---|---|---|---|---|
| 1 | George Bush | 1924 | 94 | 1133 | 34,504 |
| 2 | Gerald Ford | 1913 | 93 | 1121 | 34,133 |
| 3 | Ronald Reagan | 1911 | 93 | 1119 | 34,088 |
| 4 | John Adams | 1735 | 90 | 1088 | 33,119 |
| 5 | Herbert Hoover | 1874 | 90 | 1082 | 32,943 |
| 6 | Harry S. Truman | 1884 | 88 | 1063 | 32,373 |
| 7 | James Madison | 1751 | 85 | 1023 | 31,150 |
| 8 | Thomas Jefferson | 1743 | 83 | 998 | 30,397 |
| 9 | Richard Nixon | 1913 | 81 | 975 | 29,688 |
| 10 | John Quincy Adams | 1767 | 80 | 967 | 29,446 |

## Top 10 shortest-lived presidents

| rank | PRESIDENT | year_of_birth | lived_years | lived_months | lived_days |
|---|---|---|---|---|---|
| 1 | John F. Kennedy | 1917 | 46 | 557 | 16,978 |
| 2 | James A. Garfield | 1831 | 49 | 598 | 18,202 |
| 3 | James K. Polk | 1795 | 53 | 643 | 19,583 |
| 4 | Abraham Lincoln | 1809 | 56 | 674 | 20,516 |
| 5 | Chester Arthur | 1829 | 57 | 685 | 20,863 |
| 6 | Warren G. Harding | 1865 | 57 | 693 | 21,091 |
| 7 | William McKinley | 1843 | 58 | 703 | 21,412 |
| 8 | Theodore Roosevelt | 1858 | 60 | 722 | 21,985 |
| 9 | Calvin Coolidge | 1872 | 60 | 726 | 22,099 |
| 10 | Franklin Roosevelt | 1882 | 63 | 758 | 23,082 |

## Statistics of lived_days

| Measure | lived_days | years |
|---|---|---|
| Mean | 26,097.2 | 71.5 |
| Weighted mean (recent presidents count more) | 26,221.4 | 71.8 |
| Median | 25,673.0 | 70.3 |
| Mode | none (all values unique) | — |
| Max | 34,504.0 | 94.5 |
| Min | 16,978.0 | 46.5 |
| Standard deviation | 4,596.1 | 12.6 |




A histogram shows how lifespans are spread out: the middle half of presidents died between about 63 and 80, with fewer at the very young and very old ends. The mean and median are close, so the shape is roughly symmetric.

# Conclusions

• The typical president lived about 70 to 71 years, give or take about 13 years.

• The three longest lives were George Bush (1924), Gerald Ford (1913), Ronald Reagan (1911). Over time, though, the change is small: the first 20 presidents averaged 71.0 years and the rest averaged 71.9 years.

• Lifespans vary a lot: the gap between the longest and shortest life is 48 years.

• 6 presidents had no death date in the data, so they are not included in these results.